

Test 2

January 13, 2025

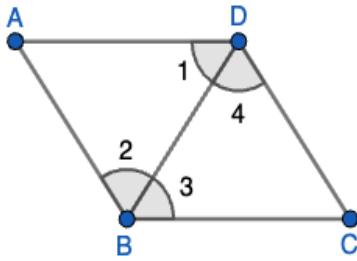
Problem 1

Determine if each of the following is true or false.

- a) If a theorem is true, its contrapositive is true.
- b) The supplement of a 40° angle is a 50° angle.
- c) If segment AB intersects segment CD at point O , then $\angle AOB \cong \angle BOD$.

Problem 2

If $\angle 1 \cong \angle 3$ and $\angle 2 \cong \angle 4$, prove that $AB \cong CD$.



Problem 3

Prove that the base angles of an isosceles triangle are congruent.

Problem 4

If l , m , and n are distinct lines such that l is parallel to m and m is parallel to n , prove that l is parallel to n .

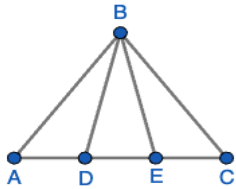
Problem 5

Determine which of the following could be the side-lengths of a triangle (mark them *yes* or *no*).

- a) 3, 4, 5.
- b) 6, 8, 15.
- c) 9, 10, 11.
- d) 5, 12, 13.
- e) 6, 8, 10.

Problem 6

If $AB \cong BC$ and $\angle ABD \cong \angle CBE$, prove that $\triangle DBE$ is isosceles.



Problem 7

Prove the following corollary to Theorem 7-1. There is at most one line perpendicular to a given line l from a point P .
[Hint: Suppose that there were two perpendiculars, and derive a contradiction]

Extra Credit

Prove that the angle sum of any triangle is a straight angle.